

Atharv C. Joshi

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Professional Summary

Motivated AI and Machine Learning Engineer with hands-on experience in autonomous robotics, computer vision, full-stack intelligent application development, and natural language processing. Proven ability to architect and ship end-to-end AI solutions — from model training to production deployment — across healthcare diagnostics, legal technology, and embedded systems domains. Consistent top performer at national-level engineering competitions (IIT Madras, IIT Bombay, COEP Pune). Proficient in Python, PyTorch, ROS2, FastAPI, React, and containerized cloud deployment pipelines.

Technical Skills

Programming Languages: Python, C, JavaScript, TypeScript, HTML, CSS

AI & Machine Learning: PyTorch, TensorFlow, OpenCV, Transfer Learning, Legal-BERT, Sentence Transformers, Computer Vision, NLP, Supervised/Unsupervised Learning

Robotics & Autonomous Systems: ROS2, Autonomous Navigation, Sensor Fusion, SLAM, Gazebo, RViz, Path Planning, Obstacle Detection

Backend & APIs: FastAPI, Uvicorn, REST APIs, Groq API (Llama-3), Google Gemini API, OpenAI GPT API, SQLAlchemy, Streamlit

Frontend: React.js, Vite, Tailwind CSS, TanStack Query, Framer Motion, Lucide-React

IoT & Embedded Systems: Raspberry Pi, Microcontrollers, ESP32, Sensor Interfacing, Real-Time Systems, Hardware Debugging

DevOps & Tools: Git, GitHub, Docker, Linux, VS Code, Vercel, Render, GitHub Pages, Tesseract OCR, Vector Databases, Cloud Dashboards

Education

D. Y. Patil Agriculture and Technical University, Kolhapur

Expected May 2027

Bachelor of Technology – Artificial Intelligence & Machine Learning

- Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Structures & Algorithms, OOP, DBMS

Padmashree Dr. Vitthalrao Vikhe Patil Sainik School, Pravaranagar

Senior Secondary (Class XII) – 63%

Nehru Montessori Senior Secondary School, Burhanpur

Secondary CBSE (Class X) – 75%

Projects

Autonomous Surveillance Rover | *Python, C, ROS2, Raspberry Pi, OpenCV, Gazebo, RViz*

- Designed and fabricated a fully autonomous ground rover using the ROS2 navigation stack integrated with on-board sensor arrays for real-time area surveillance and dynamic obstacle avoidance.
- Engineered perception modules for terrain classification, adaptive speed regulation, and real-time path re-planning using camera feeds and range sensor data in unstructured environments.
- Built end-to-end simulation workflows in Gazebo and RViz to validate navigation algorithms before hardware deployment, significantly reducing physical iteration overhead.
- Optimized ROS2 decision-making nodes for efficient edge execution on resource-constrained Raspberry Pi hardware while maintaining real-time performance requirements.

VaidyaAI – Intelligent Health OS | *React.js, FastAPI, PyTorch, OpenCV, Groq API (Llama-3), SQLAlchemy, Vercel*

- Architected a full-stack health platform delivering multi-organ clinical inference across Skin, Eye, Oral, Bone X-ray, and Lung conditions using custom PyTorch deep learning models with transfer learning.
- Developed tabular data pipelines for clinical risk stratification of Diabetes, Hypertension, and Anemia; integrated WHO Z-score analysis for pediatric malnutrition and MUAC-based image assessment.
- Built a multilingual AI wellness coach (AROMI) supporting English, Hindi, and Marathi via Groq's Llama-3 API, delivering contextual health guidance and personalized intervention recommendations.

- Implemented a location-aware hospital finder and vitals dashboard using React.js + TanStack Query with a FastAPI + SQLAlchemy backend; deployed to production on Vercel with CI/CD.

AI Legal Intelligence System (ALIS) | *FastAPI, React, TypeScript, Legal-BERT, Sentence Transformers, Tesseract OCR*

- Developed an end-to-end LegalTech platform for citizens, lawyers, and judicial officers covering case research, outcome forecasting, and automated court-document generation.
- Implemented a Case Outcome Predictor using Legal-BERT fine-tuned on 500k+ historical judgments, outputting win-probability scores alongside procedural and evidentiary risk assessments.
- Architected a semantic retrieval engine with Sentence Transformers and a Vector Database indexing 15M+ Supreme Court records and BNS 2023 statutes; all responses grounded in primary legal citations.
- Built a multilingual document drafter (6+ languages) generating ready-to-file FIRs, RTIs, and Legal Notices; integrated Tesseract OCR for parsing and explaining uploaded legal documents.
- Containerized the system with Docker; deployed FastAPI backend on Render and Vite + React + TypeScript frontend on Vercel, establishing a scalable, production-grade microservices architecture.

SehatSaathi – Multi-Condition AI Diagnostic Platform | *Python, PyTorch, OpenCV, Streamlit, OpenAI GPT API*

- Constructed a disease-detection pipeline across five clinical domains using convolutional models with transfer learning; reduced false-positive rate by 20%+ through region-of-interest preprocessing in OpenCV.
- Integrated OpenAI GPT API to power a personalized AI health coach and automated workout/nutrition planners generating weekly schedules tailored to individual fitness goals and dietary profiles.

IoT-Based Smart Monitoring Systems | *IoT Sensors, Microcontrollers, Raspberry Pi, ESP32, Cloud Dashboards*

- Engineered distributed IoT solutions for water quality monitoring, precision smart irrigation, and laser-perimeter security using ESP32, Raspberry Pi, and custom microcontroller firmware.
- Integrated sensor arrays with cloud dashboards for real-time telemetry, event logging, and automated alert dispatch; implemented rule-based actuator control for autonomous system response.

Achievements & Awards

2nd Place – Rover Domain, MindSpark 2025 (COEP Pune): Secured 4th place overall and 2nd in Rover Domain for building an autonomous navigation rover competing against national teams.

13th Rank Nationwide – RoboRun, Shastra 2026 (IIT Madras): Ranked 13th nationally as part of Team AutoNavX, developing an autonomous maze-solving robot using IR sensors and real-time navigation algorithms.

Top 5 Finalist – AutoDev Hackathon, Techfest 2025 (IIT Bombay): Selected among top 5 teams nationally in an autonomous systems engineering hackathon out of hundreds of applicants.

Cyber Suraksha Warrior of the Year 2025 – Quick Heal Foundation: Recognized for outstanding contributions to cybersecurity awareness and digital safety initiatives at a national level.

Core Competencies

Team Leadership • Analytical Problem Solving • Technical Communication • System Design • Time Management • Cross-functional Collaboration • Research & Development • Adaptability

Languages

English: Professional Proficiency

Hindi: Native

Marathi: Native